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14 July 1961
Superseding
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1 December 1960

MILITARY SPECIFICATION

BOMB, PRACTICE, PROFILE BDU-8/B

1. SCOPE. This specification establishes the requirements for one type of practice bomb designated as Bomb, Practice, Profile BDU-8/B hereafter referred to as the practice bomb.

2. APPLICABLE DOCUMENTS. The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified by this specification:

SPECIFICATIONS

Federal

TT-T-266	Thinner; Dope and Lacquer (Cellulose-Nitrate)
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof - For Packaging and Sealing

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods Of
MIL-B-131	Barrier Material; Water Vaporproof, Flexible
MIL-C-132	Crates; Open, Wood, Maximum Capacity 2500 Pounds
MIL-V-3083	Vibrator, Concrete, Pneumatic
MIL-T-9107	Test Reports, Preparation Of
MIL-L-11195	Lacquer Lustreless, Hot Spray
MIL-B-27000	Bombs, Practice, General Specification For
MIL-E-27005	Explosive Kit, Parachute Release MLU-5/B Explosive Kit, Parachute Release MLU-7/B

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

FSC 1115

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DRAWINGS

Air Force

58J46969	Bomb, Practice, Profile BDU-8/B
58D46970	Nose Section, Short
58D46971	Nose Section, Long
58D46972	Center Section Assembly
58J46973	Aft Section Assembly

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION. This specification makes provisions for preproduction testing.

3.2 COMPONENTS. The practice bomb shall consist of the following major components:

a. One Nose Section, Long, drawing 58D46971.
b. One Nose Section, Short, drawing 58D46970.
c. One Center Section Assembly, drawing 58D46972.
d. One Aft Section Assembly, drawing 58J46973.

3.3 GENERAL SPECIFICATION. The requirements of MIL-B-27000 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.4 DESIGN AND CONSTRUCTION. Design and construction of the practice bomb shall be in conformance with drawing 58J46969 and subsidiary drawings.

3.4.1 STATIC LOADS. The practice bomb shall be capable of withstanding the static loads resulting from suspension and handling, and section moment and tail fin loading.

3.4.2 CONCRETE FILL. The center section assembly, drawing 58D46972, shall be filled with concrete having a density of 143 pounds per cubic foot, plus 7, minus 18 lb/ft^3.

NOTE: The concrete fill process shall be performed by the contractor.

3.4.2.1 RECOMMENDED PROCEDURE FOR MIXING. The procedure for mixing the concrete shall be in accordance with MIL-B-27000.

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3.4.2.2 METHOD OF FILLING WITH CONCRETE. The center section of the bomb shall be filled with the concrete mixture in the following manner:

a. Place the center section in a vertical position.

b. Fill with concrete mixture to one quarter of the distance of the full mark.

c. Insert the pneumatic vibrator conforming to MIL-V-3083 as shown in figure 1 and vibrate the mixture for a period of 45 ± 15 seconds. At the end of this step, remove the vibrator.

d. Fill to one half of the distance of the full mark and repeat step c.

e. Fill to three quarters of the distance of the full mark and repeat step c.

f. Fill the center section to the full mark with the concrete mixture and repeat step c.

3.4.2.3 SETTING AND CURING TIME. Setting and curing times shall be in accordance with MIL-B-27000.

3.5 FINISH. The finish shall be as specified in MIL-B-27000.

3.5.1 FINISH COAT. All metal parts except those which constitute internal bomb parts shall be given a finish coat in accordance with the following:

a. Lacquer, Lustreless, MIL-L-11195, or equivalent, Color No. 3714, shall be used for the finish coat.

b. The lacquer shall be thinned by one volume of lacquer thinner, TT-T-266, to two volumes of lacquer.

c. The lacquer shall be applied by spraying on a coating of 0.5 to 1.0 mil in thickness.

d. Parts shall be allowed to air dry for a minimum of one hour prior to handling after receiving the finish coat.

3.6 PERFORMANCE. For the purpose of this specification, satisfactory performance shall be the initiation of the Cartridge, Actuating Device MLU-4/B or the MLU-6/B.

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1. Fill Mark
2. Concrete Mixture
3. Concrete Vibrator

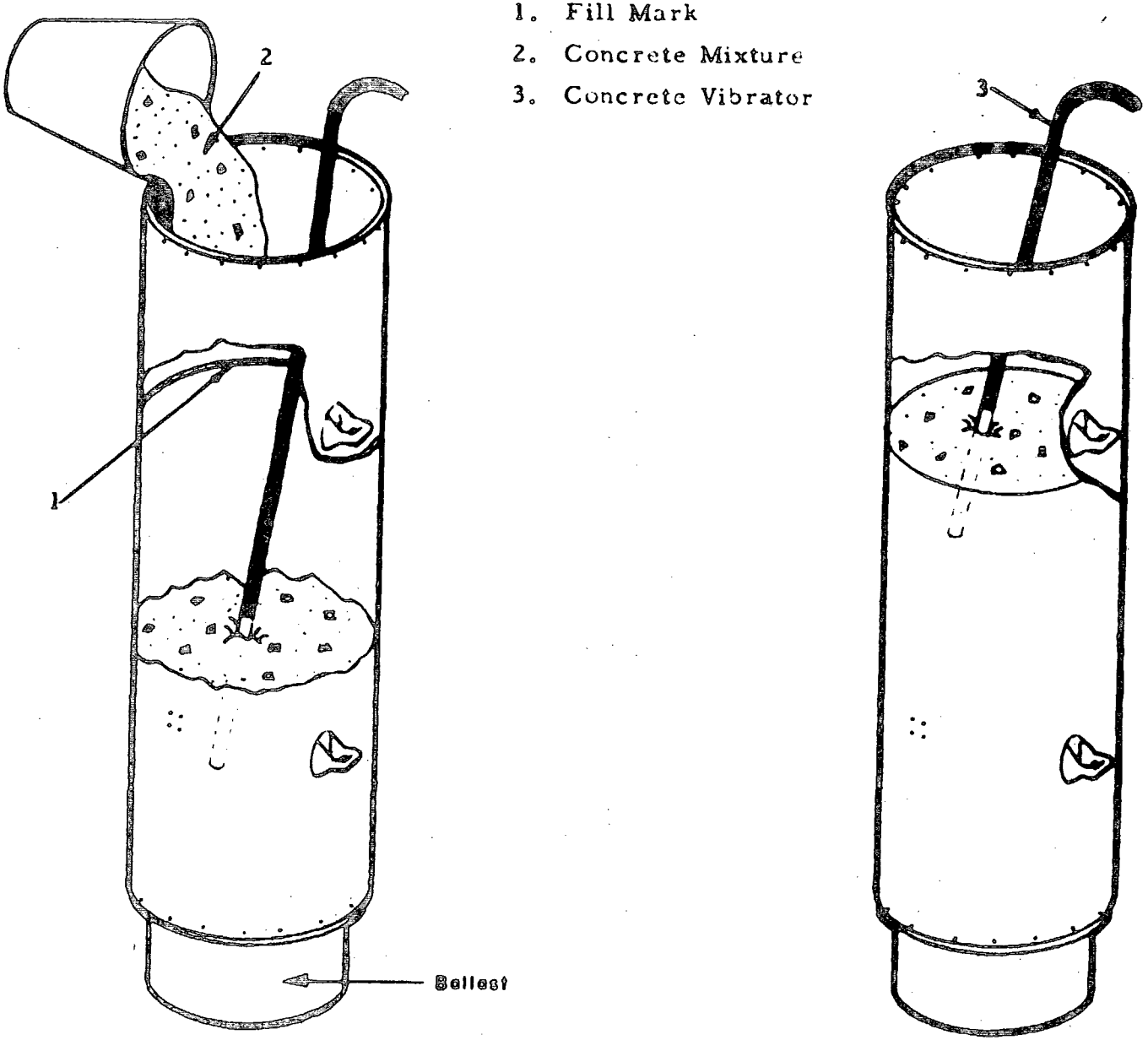


Figure 1. Concrete Fill Process Bomb, Practice, Profile, BDU-8/B

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3.7 SHOCK. The practice bomb shall be capable of meeting the performance requirements after being subjected to the shock loads specified in section "4.5 TEST METHODS."

3.8 VIBRATION. The practice bomb shall be capable of meeting the performance requirements after being subjected to the vibrations specified in "4.5 TEST METHODS."

3.9 GOVERNMENT FURNISHED PROPERTY. When the contract or purchase order so provides, the Government will furnish the following to the contractor, upon his request, for use in conducting the static, shock, and vibration tests; or for insertion of the primacord into assembled bombs. See 6.2.2.

Item No.	Drawing No.	Item Description	Quantity
1	58D6362	Parachute System, Cargo, 16 Foot Diameter, Heavy	See 6.2.2
2	58A45989	Explosive Kit, Parachute Release MLU-5/B	See 6.2.2
or			
2	58A45990	Explosive Kit, Parachute Release MLU-7/B	See 6.2.2
3	58B45977-5	Test Cartridge	See 6.2.2

3.10 IDENTIFICATION OF PRODUCT. Equipment assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.11 WORKMANSHIP. Workmanship shall be in accordance with MIL-B-27000.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS. The inspection and testing of the Bomb, Practice, Profile BDU-8/B shall be classified as follows:

a. Preproduction Tests
b. Acceptance Tests

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES. The preproduction test samples shall consist of units representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

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4.2.2 PREPRODUCTION TEST REPORT. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.2.3 PREPRODUCTION TESTS. Preproduction tests shall consist of all tests described under section "4.5 TEST METHODS" and shall be performed in the same order.

4.3 ACCEPTANCE TESTS. Acceptance tests shall consist of the following individual tests:

a. Examination of Product
b. Conformance to Drawings
c. Firing Test of Initiator Trigger Assembly

4.4 TEST CONDITIONS. Test conditions shall be as required for each specific test.

4.5 TEST METHODS

4.5.1 EXAMINATION OF PRODUCT. Each practice bomb shall be given a complete visual examination to determine compliance with the requirements of this specification, including workmanship, materials, and correctness of assembly.

4.5.1.1 Insofar as possible, inspection shall be done after the bomb is completely assembled.

4.5.1.2 INSPECTION OF INITIATOR TRIGGER ASSEMBLY. Prior to installation into the bomb, unscrew the trigger housing, part No. 58B45984, from the initiator trigger assembly, and examine for visible DC-6 Compound. All visible DC-6 Compound shall be wiped off with a clean cloth.

4.5.2 CONFORMANCE TO DRAWINGS. Each practice bomb shall be inspected for conformance to the engineering drawings, and to all specifications and standards referenced therein.

4.5.3 FIRING TEST OF INITIATOR TRIGGER ASSEMBLY

a. Assemble parts Nos. 58B45979 Booster Sleeve, Initiator, 58C45982 Initiator Housing, and 58B45977-5 as shown on drawing 59C46000, except that Part No. 58C45982 shall not be staked. This assembly shall hereafter be referred to as "test cartridge."

b. Assemble the initiator trigger, part No. 58B46977 to the connection ring, part No. 58C46997.

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c. Assemble test cartridge into the connection ring. In lieu of staking part No. 58C45982, use any suitable means for retaining part No. 58B46977-5 in the initiator housing. Use of a shim is one satisfactory method.

d. Manually pull the trigger release. Firing indicates that the Initiator Trigger Assembly is acceptable. Failure to fire indicates out-of-tolerance parts or improper assembly, and the Initiator Trigger Assembly shall be rejected.

4.5.4 STATIC LOAD TESTS. All static load tests shall be conducted with the retarding parachute, and either the MLU-5/B or the MLU-7/B Explosive Kit installed in the bomb.

4.5.4.1 REFERENCE POINT. Reference point for locating stations for static loads shall be: Station 0, forward surface of long nose section.

4.5.4.2 SUSPENSION TEST, PROCEDURE I

4.5.4.2.1 TEST SET-UP. The test set-up shall be as described in procedure I, MIL-B-27000.

4.5.4.2.2 METHOD OF APPLYING LONGITUDINAL LOADS. Longitudinal loads shall be applied directly to the main mass of the bomb through the handling lugs.

4.5.4.2.3 TEST PROCEDURE. The test procedure shall be in accordance with procedure I of MIL-B-27000. Design load and direction of application shall be as follows:

a. Case I

(1) In the vertical plane:

Station	54.5 down	5803 pounds
	91.0 down	6660 pounds
	142.38 down	1645 pounds

(2) In the horizontal plane:

Station	37.0 left	1389 pounds
	61.25 right	5192 pounds
	85.25 right	5192 pounds
	100.00 right	1017 pounds

(3) In the fore and aft plane at each handling lug:

forward	808 pounds

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b. Case II

(1) In the vertical plane:

Station	54.5 down	5803 pounds
	91.0 down	6660 pounds
	141.375 up	3250 pounds

(2) In the horizontal plane:

Station	27.725 right	1693 pounds
	61.25 right	5192 pounds
	85.25 right	5192 pounds

(3) In the fore and aft plane at each handling lug:

aft	3354 pounds

c. Case III

(1) In the vertical plane:

Station	54.5 down	5803 pounds
	91.0 down	6660 pounds
	141.375 up	1500 pounds

(2) In the horizontal plane:

Station	61.25 right	5192 pounds
	85.25 right	5192 pounds
	99.00 right	4000 pounds

(3) In the fore and aft plane at each handling lug:

aft	3354 pounds

4.5.4.3 SUSPENSION TEST, PROCEDURE II

4.5.4.3.1 TEST SET-UP. The test set-up shall be as follows:

a. Case I: The test set-up shall be in accordance with procedure II of MIL-B-27000 except that the additional downward force shall be applied at station 77.5.

b. Case II: The test set-up shall be in accordance with procedure III of MIL-B-27000, except that the additional downward forces shall be applied at stations 56.5 and 92.5.

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4.5.4.3.2 TEST PROCEDURES. The test procedures shall be as follows:

a. Case I: Test procedures shall be in accordance with procedure II of MIL-B-27000 except that the downward forces shall be applied at station 77.5.

b. Case II: Test procedures shall be in accordance with procedure III of MIL-B-27000 except that the downward forces shall be applied at stations 56.5 and 92.5.

4.5.4.4 FORE AND AFT SECTION MOMENT TEST

4.5.4.4.1 TEST SET-UP. The test set-up shall be in accordance with procedure I of the suspension test of MIL-B-27000 except that cradles shall be placed under the bomb at stations 57.00 and 91.50. The cradles shall be capable of withstanding the applied loads and preventing imposition of the loads on the suspension lugs.

4.5.4.4.2 TEST PROCEDURES

4.5.4.4.2.1 MAIN BALLAST ATTACHMENT. The nose assembly shall be removed. A downward load of 12,505 pounds shall be applied at station 37.37. This force shall be applied in increments of 0, 50, 66.7, 75, 85, 90, 95 and 100 percent of the full load.

Evidence of permanent deformation during loading up to 90 percent of full load or failure during loading up to full load shall be cause for rejection.

4.5.4.4.2.2 AFT SECTION. Apply a downward load of 8423 pounds at station 137.3. This force shall be applied in increments of 0, 50, 66.7, 75, 85, 90 and 100 percent of full load.

Evidence of permanent deformation during loading up to 90 percent of full load or failure during loading up to full load shall be cause for rejection.

4.5.4.5 FIN TEST

4.5.4.5.1 TEST SET-UP. The test set-up for the fin test shall be as follows:

a. The aft section assembly shall be attached to a rigid frame capable of withstanding the test loads. Suitable means shall be provided to measure fin deflection in the direction of load application.

4.5.4.5.2 TEST PROCEDURES. Concentrated loads shall be applied simultaneously perpendicular to the surface of any two fins at the

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center of pressure, as indicated on figure 2, provided that one load is clockwise and the other load is counter-clockwise. Sequence and load values shall be as follows:

a. Preload each fin to 400 pounds, and establish a zero reference.

b. Increase the test load directly to 600 pounds.

c. Increase the load in increments of 75 pounds to a total of 1500 pounds. Fin deflection shall be measured at each 75 pound increment up to and including 1500 pounds.

d. Reduce the loads simultaneously to 400 pounds.

e. Determine whether permanent deformation exists.

f. Increase the load directly to 1690 pounds.

g. Increase the load by increments of 50 pounds to a total of 2190 pounds. Fin deflection shall be measured at each 50 pound increment.

h. Reduce loads simultaneously to 400 pounds.

i. Determine whether permanent deformation exists, or failure occurs.

Permanent deformation after loading to 1500 pounds or failure while loading to 2190 pounds shall constitute failure to pass the test.

4.5.5 FUNCTIONAL TEST. Install a Cartridge, Actuating Device MJU-4/B or MJU-6/B in the bomb. Actuate the pullout and firing mechanism. Failure to initiate the cartridge shall be cause for rejection.

NOTE: During this test provisions shall be made for protection against flash damage and cartridge case fragmentation.

4.5.6 SHOCK TEST. The representative bomb, with concrete fill as specified in 3.4.2, and retarding parachute and either the MJU-5/B or MJU-7/B Explosive Kit installed, shall be subjected to the Shock Test specified in MIL-B-27000.

Firing of any explosive component, structural failure, or cracking of concrete fill during this test shall be cause for rejection.

4.5.7 VIBRATION TEST. The representative bomb, with concrete fill as specified in 3.4.2, and retarding parachute and either the MJU-5/B

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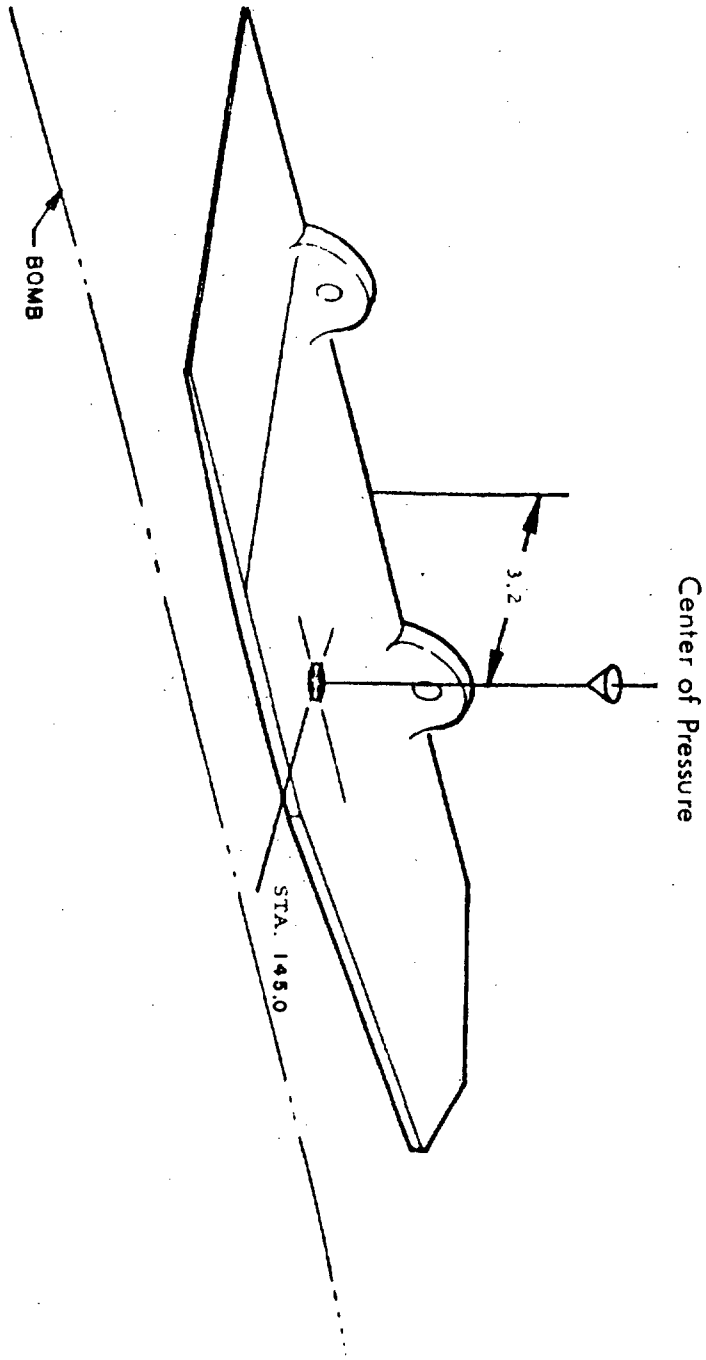


Figure 2. Application of Concentrated Loads

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or MJU-7/B Explosive Kit installed, shall be subjected to the Vibration Test specified in MIL-B-27000.

Firing of any explosive component, structural failure, or cracking of concrete fill during this test shall be cause for rejection.

NOTE: During this test proper barricade shall be utilized to prevent damage to adjacent equipment or injury of personnel in the event of initiation of any explosive component.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION, ALL LEVELS, BOMB ASSEMBLED OR DISASSEMBLED. All machined surfaces not having a protective finish shall be preserved in accordance with method I, MIL-P-116. Disassembled bombs, and assembled bombs with or without the parachute, shall have all openings covered with barrier material conforming to MIL-B-131, and sealed with tape conforming to PPP-T-60. Openings in the bomb include, but are not limited to the following:

a. Screw holes with or without screws in place.

b. Opening for safety pin.

c. Peripheral separations.

d. All large openings when bomb is shipped disassembled.

5.2 PACKAGING AND PACKING, ALL LEVELS. The bomb or components as designated by the procuring activity, shall be packed in a crate conforming to MIL-C-132. The fins and attaching hardware shall be removed from the bomb and secured inside the crate. Physical protection shall be in accordance with JAN-P-100.

5.3 MARKING. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

BOMB, PRACTICE, PROFILE BDU-8/B

6. NOTES

6.1 INTENDED USE. The Bomb, Practice, Profile BDU-8/B is intended to be used to provide practice in aircraft maneuverability and cruise control while simulating carriage of the parent weapon and to provide practice in delivery by duplicating the trajectory of the parent weapon. The BDU-8/B Bomb is a reuseable practice weapon.

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6.1.1 REUSE CRITERIA. After the bomb has been dropped, it will be suitable for reuse if it has stuck into the earth without falling over or otherwise inflicting damage to the aft section.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Conditions under which Government furnished property will be made available to the contractor.

c. Whether preproduction tests will be conducted by the procuring activity.

d. Whether bomb shall be shipped assembled or disassembled.

6.2.1 PREPRODUCTION TEST SAMPLES. It is expected that the contract or purchase order will specify that a minimum of four units representative of the production practice bomb will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of inspection for these tests.

6.2.2 QUANTITIES OF GOVERNMENT FURNISHED PROPERTY. Quantities of items 1, 2 and 3 listed in 3.9 shall be specified in the contract by the procuring activity. Factors to be considered are:

a. Whether retardation parachute is to be shipped in assembled bombs or shipped separately.

b. Whether the primacord assembly is to be inserted into assembled bombs prior to shipment, or is to be shipped as part of the MLU-5/B or MLU-7/B Explosive Kit. See also section 5 of MIL-E-27005.

c. Number of explosive kits considered essential for use in the static, shock, and vibration tests.

d. The requirements set forth in 6.2.3 for the firing pin test of 4.5.3.

6.2.3 QUANTITIES OF PARTS REQUIRED FOR THE FIRING TEST OF INITIATOR TRIGGER ASSEMBLY

a. Part No. 58B45977-5 should be ordered in a ratio of 11 parts per 10 bombs.

b. Parts Nos. 58B45979, Booster Sleeve, Initiator, and 58C45982, Initiator Housing, should be ordered in a ratio of one each per 50 bombs.

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c. Part No. 58C46997 will be furnished by the bomb contractor and one only is necessary for all tests.

6.3 PRODUCTION DELIVERY. No production item should be delivered prior to approval of preproduction samples.

6.4 DEFINITIONS

a. PREPRODUCTION TESTS. Preproduction tests are those tests performed prior to production on representative models of the production bomb to demonstrate that the equipment meets the requirements of this specification.

b. ACCEPTANCE TESTS. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the bombs are equivalent in performance and construction to the approved preproduction samples.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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